Mock Exam 2 — Solutions

Lectures 5–8 (Chapters 4–6)

Prof. Dr. Christoph Weisser

Quantitative Research Methods · HSBI

Summer Semester 2026

Overview — problems and points

Problem	Topic	Points
P1	Logistic regression by hand	18
P2	Discriminant analysis and naive Bayes — concepts	12
P3	Confusion matrix and ROC curve	15
P4	Resampling methods (validation set · CV · bootstrap)	18
P5	Model selection with C_p and BIC	12
P6	Ridge regression and the lasso	15
	Total (90 minutes; points $\approx$ minutes)	**90**

How to use this deck

Attempt each problem under exam conditions before looking at the teal boxes. Allowed aids in the exam: a non-programmable calculator and one handwritten A4 sheet of notes (both sides). Round final results to 3 decimals.

Problem 1 — Logistic regression by hand (18 P)

A bank models the default probability of a credit-card customer as a function of the monthly balance x (in EUR; observed range 0–400):

$$\hat{p}(x) = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1 x}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1 x}}, \qquad \hat{\beta}_0 = -4, \quad \hat{\beta}_1 = 0.02.$$

(a) Predicted default probability at $x = 100$ and $x = 250$. (4 P)

(b) Odds and log-odds at both balances; interpret the odds at $x = 250$. (4 P)

(c) Balance x^* with $\hat{p}(x^*) = 0.5$. (3 P)

(d) Factor change of the odds per EUR and per 100 EUR; why is "probability $+0.02$ per EUR" wrong? (4 P)

(e) Two reasons why linear regression on the 0/1 response is inappropriate. (3 P)

Problem 1 — solution (a) and (b)

Solution 1(a) — predicted probabilities

Linear predictor (log-odds): $\hat{\eta}(x) = -4 + 0.02\,x$, so $\hat{\eta}(100) = -4 + 2 = -2$ and $\hat{\eta}(250) = -4 + 5 = 1$. Hence

$$\hat{p}(100) = \frac{e^{-2}}{1 + e^{-2}} = \frac{0.1353}{1.1353} = 0.119, \qquad \hat{p}(250) = \frac{e^{1}}{1 + e^{1}} = \frac{2.7183}{3.7183} = 0.731.$$

Solution 1(b) — odds and log-odds

Odds $= \hat{p}/(1 - \hat{p}) = e^{\hat{\eta}(x)}$ and log-odds $= \hat{\eta}(x)$:

	$x = 100$	$x = 250$
log-odds	-2.000	1.000
odds	$e^{-2} = 0.135$	$e^{1} = 2.718$

Check with (a): $0.119/0.881 = 0.135$ and $0.731/0.269 = 2.718$. ✓ At $x = 250$ the odds are about 2.72 to 1: default is roughly 2.7 times as probable as non-default.

Problem 1 — solution (c) and (d)

Solution 1(c) — balance with a 50:50 prediction

$\hat{p} = 0.5 \iff \text{odds} = 1 \iff \text{log-odds} = 0$:

$$-4 + 0.02\,x^* = 0 \quad\Longrightarrow\quad x^* = \frac{4}{0.02} = 200\ \text{EUR}.$$

Below 200 EUR "no default" is predicted as more likely; above 200 EUR "default".

Solution 1(d) — interpretation on the odds scale

Per additional EUR the odds are *multiplied* by $e^{\hat{\beta}_1} = e^{0.02} = 1.020$ ($+2.0\,\%$); per additional 100 EUR by $e^{100\cdot 0.02} = e^2 = 7.389$. The quoted statement is wrong because $\hat{\beta}_1$ acts on the *log-odds*: the probability change per EUR is $\partial\hat{p}/\partial x = \hat{\beta}_1\,\hat{p}(x)\big(1-\hat{p}(x)\big)$, which depends on x — the S-curve is steepest at $\hat{p} = 0.5$ and almost flat in the tails, so there is no constant probability effect.

Problem 1 — solution (e)

Solution 1(e) — why not linear regression

(1) The straight line $\hat{\beta}_0 + \hat{\beta}_1 x$ is unbounded: for small or large balances it returns "probabilities" below 0 or above 1 — not sensible estimates of $\Pr(Y = 1 \mid x)$. The logistic function always stays in $[0, 1]$.

(2) A 0/1 response violates the linear-model error assumptions: errors are non-normal and heteroscedastic — $\mathrm{Var}(Y \mid x) = p(x)(1 - p(x))$ depends on x — so the usual inference is unreliable.

(With > 2 classes a numeric coding would additionally impose an artificial ordering and equal spacing between the classes.)

Remember

Logistic regression is a *linear model for the log-odds*: $\log\big(p/(1-p)\big) = \beta_0 + \beta_1 x$. Coefficients are additive on the log-odds scale and multiplicative on the odds scale — never additive on the probability scale.

Problem 2 — restatement

Problem 2 — Discriminant analysis and naive Bayes (12 P)

(a) LDA vs. QDA: the distinguishing covariance assumption; number of covariance matrices estimated with K classes. (4 P)

(b) Bias–variance: when does LDA beat QDA on test data and vice versa? Role of n and of the true class covariances. (3 P)

(c) Key assumption of naive Bayes; why can it help when p is large relative to n? (3 P)

(d) Which classifiers give linear and which give quadratic or more general boundaries? (2 P)

Problem 2 — solution (a) and (b)

Solution 2(a) — covariance assumptions

Both assume $X \mid Y = k \sim \mathcal{N}(\mu_k, \Sigma_k)$. **LDA:** one covariance matrix *shared* by all classes ($\Sigma_1 = \cdots = \Sigma_K = \Sigma$) $\Rightarrow$ estimates **1** matrix ($p(p+1)/2$ parameters). **QDA:** a *class-specific* Σ_k $\Rightarrow$ estimates **K** matrices ($K\,p(p+1)/2$ parameters).

Solution 2(b) — who wins when

QDA is far more flexible: *low bias but high variance*; LDA is restrictive: *low variance but biased if the* Σ_k *truly differ*.

- **LDA wins** when n is small (or p large) — variance reduction is crucial — and when the class covariances are approximately equal (little bias incurred).
- **QDA wins** when n is large — its many parameters are estimated reliably — and when the class covariances clearly differ, so the shared-Σ assumption of LDA would cause substantial bias.

Problem 2 — solution (c) and (d)

Solution 2(c) — naive Bayes

Assumption: *within each class the p predictors are independent*: $f_k(x) = f_{k1}(x_1) \cdots f_{kp}(x_p)$. Only p one-dimensional marginals per class are estimated instead of a full p-dimensional joint density (no within-class association parameters). This slashes the number of parameters and hence the *variance*; for large p relative to n the variance saving usually outweighs the bias from ignoring dependence — the bias–variance trade-off again.

Solution 2(d) — boundary shapes

LDA: linear boundaries (quadratic terms cancel due to the shared Σ). **QDA:** quadratic boundaries. **Naive Bayes:** generally non-linear but *additive* boundaries — log-odds of the form $\sum_j g_j(x_j)$ without interactions; only in special cases (e.g. Gaussian marginals with class-shared variances) does it become linear.

Problem 3 — restatement

Problem 3 — Confusion matrix and ROC curve (15 P)

Logistic-regression default classifier on $n = 1{,}000$ customers with threshold $\hat{p} > 0.5$:

		True status	
		Default	No default
Predicted	Default	60	45
	No default	40	855
	Total	100	900

(a) TP/FP/FN/TN; accuracy and error rate; sensitivity; specificity; precision; FPR; interpret sensitivity and precision. (7 P)

(b) Effect of lowering the threshold from 0.5 to 0.2 on sensitivity and specificity; business reason for accepting it. (4 P)

(c) What does the ROC curve plot? Meaning of AUC $= 0.5$ and AUC $= 1$. (4 P)

Problem 3 — solution (a)

Solution 3(a) — the five metrics

Positive class "default": TP $= 60$, FP $= 45$, FN $= 40$, TN $= 855$.

accuracy	$(60+855)/1000$	$= 0.915$	error rate $= 0.085$
sensitivity (recall/TPR)	$60/(60+40) = 60/100$	$= 0.600$	
specificity	$855/(855+45) = 855/900$	$= 0.950$	
precision	$60/(60+45) = 60/105$	$= 0.571$	
false positive rate	$45/(45+855) = 45/900$	$= 0.050$	$= 1 -$ specificity

Sensitivity: 60 % of the true defaulters are detected — 40 % are missed. *Precision:* of all customers flagged as defaulters only 57.1 % actually default; the remaining flags are false alarms.

Problem 3 — solution (b) and (c)

Solution 3(b) — lowering the threshold to 0.2

More customers are flagged: true defaulters with $0.2 < \hat{p} \leq 0.5$ are now caught $\Rightarrow$ **sensitivity increases** (FN $\downarrow$); more non-defaulters are flagged too $\Rightarrow$ **specificity decreases** (FPR $\uparrow$). *Business reason:* asymmetric costs — a missed default (credit loss) is far more expensive than a false alarm (e.g. an unnecessary review); the expected cost can fall even if the raw error rate rises.

Solution 3(c) — ROC and AUC

The ROC curve plots the *true positive rate* (sensitivity) against the *false positive rate* ($1-$specificity) as the classification *threshold* varies from 1 down to 0. AUC $= 0.5$: diagonal — no better than random guessing. AUC $= 1$: perfect classifier — some threshold attains sensitivity $= 1$ and specificity $= 1$ simultaneously. (AUC $=$ probability that a random defaulter is scored above a random non-defaulter.)

Problem 4 — Resampling methods (18 P)

(a) $n = 200$: number of model *fits* for the validation-set approach (50/50 split) · 10-fold CV · LOOCV — with justification. (4 P)

(b) LOOCV vs. 10-fold CV: bias and variance; which is preferred in practice? (4 P)

(c) Show $\Pr(i \in \text{bootstrap sample}) = 1 - (1 - 1/n)^n$; evaluate at $n = 5$; limit as $n \to \infty$. (5 P)

(d) $B = 3$ bootstrap estimates $\hat{\alpha}^* = 0.4$ / 0.5 / 0.6 (toy example; in practice $B \approx 1{,}000$): compute $\widehat{\mathrm{SE}}_B(\hat{\alpha})$; what does it estimate; why is the bootstrap useful? (5 P)

Problem 4 — solution (a) and (b)

Solution 4(a) — number of fits at n

- **Validation set: 1 fit** — train once on the 100 training observations, evaluate on the 100 held-out ones.
- **10-fold CV: 10 fits** — each fold of 20 is held out once; refit on the remaining 180 each time.
- **LOOCV: 200 fits** — each observation is held out once; refit on the other 199 (= n-fold CV; for least squares a shortcut formula exists).

Solution 4(b) — bias and variance of CV

Bias: LOOCV trains on $n-1$ observations $\Rightarrow$ nearly unbiased; 10-fold trains on $0.9\,n$ $\Rightarrow$ slightly overestimates the test error. *Variance:* LOOCV is *higher*: its n fits use almost identical training sets, so the fold errors are strongly positively correlated — averaging highly correlated quantities reduces variance little; the 10 folds overlap less. *Practice:* $k = 5$ or 10 — good bias–variance compromise and far cheaper.

Problem 4 — solution (c) and (d)

Solution 4(c) — bootstrap inclusion probability

Each of the n draws (with replacement) misses observation i with probability $1-1/n$; draws are independent: $\Pr(i \notin \text{sample}) = (1-1/n)^n \Rightarrow \Pr(i \in \text{sample}) = 1-(1-1/n)^n$.

$$n = 5: \;\; 1-(4/5)^5 = 1-0.32768 = 0.672; \qquad n \to \infty: \;\; 1-e^{-1} \approx 1-0.368 = 0.632.$$

A bootstrap sample contains roughly 2/3 of the distinct original observations.

Solution 4(d) — bootstrap standard error

$\bar{\alpha}^* = (0.4+0.5+0.6)/3 = 0.5;$

$$\widehat{\mathrm{SE}}_B(\hat{\alpha}) = \sqrt{\tfrac{1}{3-1}\left[(-0.1)^2 + 0^2 + (0.1)^2\right]} = \sqrt{0.02/2} = \sqrt{0.01} = 0.100.$$

It estimates how much $\hat{\alpha}$ would vary over repeated samples from the population. The bootstrap needs no analytic formula and no new data — it resamples the observed data and works for almost any statistic.

Problem 5 — Model selection with Cp and BIC (12 P)

Best subset selection with $n = 100$ and $\hat{\sigma}^2 = 4$ (from the full model):

d	1	2	3	4
RSS_d	500	444	428	424

$$C_p = \frac{1}{n}\left(\mathrm{RSS} + 2d\hat{\sigma}^2\right), \qquad \mathrm{BIC} = \frac{1}{n}\left(\mathrm{RSS} + \log(n)\, d\, \hat{\sigma}^2\right), \qquad \log(100) \approx 4.605.$$

(a) C_p for all four models. (4 P)

(b) BIC for all four models. (4 P)

(c) Which d does each criterion select? Explain the disagreement. (2 P)

(d) Why can training RSS alone never select d? (2 P)

Problem 5 — solution (a) and (b)

Solution 5(a) — Cp (penalty 8 per predictor)

$C_p = (\mathrm{RSS}_d + 2d\hat{\sigma}^2)/100 = (\mathrm{RSS}_d + 8d)/100$:

d	1	2	3	4
$\mathrm{RSS}_d + 8d$	508	460	452	456
C_p	5.080	4.600	**4.520**	4.560

Solution 5(b) — BIC (penalty 18.421 per predictor)

$\mathrm{BIC} = (\mathrm{RSS}_d + 4.605 \cdot 4 \cdot d)/100 = (\mathrm{RSS}_d + 18.421\, d)/100$:

d	1	2	3	4
$\mathrm{RSS}_d + 18.421\, d$	518.421	480.841	483.262	497.683
BIC	5.184	**4.808**	4.833	4.977

Problem 5 — solution (c) and (d)

Solution 5(c) — selected models and why they differ

C_p picks d $= 3$ (4.520); BIC picks d $= 2$ (4.808). Both are RSS + size penalty, but BIC charges $\log(n)\hat{\sigma}^2 = 18.421$ per predictor vs. $2\hat{\sigma}^2 = 8$ for C_p; since $\log(100) = 4.605 > 2$, BIC penalises extra predictors more heavily and selects smaller models. Here the RSS drop from $d = 2$ to $d = 3$ is 16: larger than 8 (so C_p accepts the third predictor) but smaller than 18.421 (so BIC rejects it).

Solution 5(d) — why RSS alone fails

Adding a predictor can never increase the training RSS: RSS decreases *monotonically* in d (training R^2 increases monotonically), so RSS-based selection would always choose the largest model $d = 4$ — training error is an ever more optimistic estimate of test error as flexibility grows. Hence: penalise model size (C_p / AIC / BIC / adjusted R^2) or estimate test error directly (validation set / cross-validation).

Problem 6 — Ridge regression and the lasso (15 P)

Linear model with standardised predictors $X_1, \ldots, X_p$.

- (a) Objectives of ridge and lasso; which coefficient is not penalised? (4 P)
- (b) Behaviour as $\lambda \to 0$ and $\lambda \to \infty$; which Chapter-3 method is recovered as $\lambda \to 0$? (3 P)
- (c) Why can the lasso set coefficients exactly to zero (geometry for $p = 2$)? (3 P)
- (d) Why standardise the predictors first? (2 P)
- (e) Ridge fits at $\lambda = 0.1$ and $\lambda = 100$ (standardised coefficients) and a 10-fold CV grid: which fit has lower bias / lower variance; which λ should be chosen? (3 P)

	$\lambda = 0.1$	$\lambda = 100$
$\hat{\beta}_1$	0.92	0.31
$\hat{\beta}_2$	0.85	0.29
$\hat{\beta}_3$	-0.41	-0.12
$\hat{\beta}_4$	0.20	0.06

λ	0.1	1	10	100
CV error	12.8	11.9	10.6	13.4

Problem 6 — solution (a) and (b)

Solution 6(a) — the two objectives

$$\text{Ridge:}\ \min_{\beta}\ \mathrm{RSS} + \lambda \sum_{j=1}^{p} \beta_j^2, \qquad \text{Lasso:}\ \min_{\beta}\ \mathrm{RSS} + \lambda \sum_{j=1}^{p} |\beta_j|,$$

with $\mathrm{RSS} = \sum_i \big(y_i - \beta_0 - \sum_j \beta_j x_{ij}\big)^2$ and tuning parameter $\lambda \geq 0$. The intercept β_0 is *not* penalised.

Solution 6(b) — limiting behaviour

$\lambda \to 0$: the penalty vanishes; both approach the *ordinary least-squares* fit of Chapter 3. $\lambda \to \infty$: the penalty dominates; all penalised coefficients go to zero (null model with intercept only). Ridge shrinks *towards* zero (exactly zero only in the limit); the lasso sets more and more coefficients *exactly* to zero as λ grows — flexibility decreases continuously along the λ path.

Problem 6 — solution (c) and (d)

Solution 6(c) — why the lasso yields exact zeros

Equivalent constrained form: minimise RSS subject to $|\beta_1| + |\beta_2| \leq s$ (lasso: a *diamond* with corners on the axes) or $\beta_1^2 + \beta_2^2 \leq s$ (ridge: a *disk* with no corners). The solution lies where the elliptical RSS contours around the OLS estimate first touch the region. Ellipses typically hit the diamond at a *corner* — where one coefficient is exactly zero (sparsity / variable selection). The disk has no corners, so the contact point generically has all coordinates non-zero: ridge shrinks but keeps every predictor.

Solution 6(d) — why standardise

OLS is scale-equivariant — rescaling X_j just rescales $\hat{\beta}_j$. The penalties depend on the coefficient *magnitudes*: a predictor in small units has a large coefficient and would be shrunk more strongly for purely cosmetic reasons. Standardising to standard deviation one makes the penalty act comparably on all predictors and the fit independent of the units.

Problem 6 — solution (e)

Solution 6(e) — bias-variance and the choice of lambda

$\lambda = 0.1$: almost no shrinkage — close to the OLS fit — more flexible $\Rightarrow$ **lower bias and higher variance**. $\lambda = 100$: strong shrinkage (e.g. $0.92 \to 0.31$; each coefficient about a third of its size) $\Rightarrow$ **higher bias and lower variance**. This is the Chapter-2 bias–variance trade-off controlled by the single knob λ. From the CV table the estimated test error is smallest at $\lambda = 10$ ($10.6 < 11.9 < 12.8 < 13.4$) — choose $\lambda = 10$; neither of the two fits in the coefficient table is optimal.

Big picture for the exam

Regularisation trades a little bias for a large variance reduction; λ is always chosen by cross-validation — never by looking at training RSS.